

# ELEC5305 Project Proposal

## 1. Project Title

**Analysis of Audio Signal Representations for Liquid Fill-Level Estimation from Pouring Sounds**

## 2. Student Information

**Full Name:** Yueyao Lyu

**Student ID (SID):** 540568813

**GitHub Username:** lyu-ui

**GitHub Pages link:** <https://lyu-ui.github.io/elec5305-project-540568813/>

**GitHub Project Link:** <https://github.com/lyu-ui/elec5305-project-540568813>

## 3. Project Overview

This project aims to investigate whether the liquid fill level of a container can be estimated from the sound produced while water is being poured into it. Pouring sounds contain time-varying acoustic information that is influenced by the amount of liquid in the container and the resonant properties of the container-air-liquid system.

The project will compare several audio signal representations and features for fill-level estimation. A baseline system will first be developed using short-time Fourier transform (STFT) analysis and conventional spectral features. As the project progresses, additional signal-processing techniques introduced during ELEC5305 will be investigated where appropriate, such as modulation-based features, auditory filter-bank representations, adaptive time-frequency processing, and possibly linear prediction. The overall research question is:

***How effectively can different audio signal-processing representations and features be used to estimate liquid fill level from pouring sounds?***

## 4. Background and Motivation

Liquid-level estimation is useful in automated monitoring, industrial systems, household devices, and robotic applications. Conventional systems often rely on dedicated sensors, cameras, or ultrasonic devices. Audio provides an alternative source of information because the physical properties of a container and its contents affect the sound produced during pouring.

Bagad et al. (2025) showed that pouring audio can be used to infer physical properties including liquid level, container dimensions, pouring rate, and time to fill. Their work demonstrates that pitch and other acoustic characteristics vary systematically as a container fills.

Other studies have investigated auditory and tactile perception for water filling-level estimation (Qian et al., 2025) and vibration-resonance-based approaches for estimating container fill level (Schmelter et al., 2020). These works suggest that resonance and spectral behaviour can provide useful information about fill level.

This project will take a signal-processing-focused approach. Rather than reproducing a complex deep-learning system, it will compare interpretable audio representations and features. This allows the relationship between acoustic behaviour and the signal-processing concepts studied in ELEC5305 to be examined directly.

## 5. Proposed Methodology

The project will be implemented primarily in MATLAB. The primary dataset will be the publicly available pouring-audio data associated with The Sound of Water study (Bagad et al., 2025), from which a controlled subset of water-pouring recordings will be selected.

The first stage will involve preprocessing. Recordings will be converted to a consistent format where necessary, normalised in amplitude, and segmented to retain the relevant pouring interval. Time-domain waveforms and frequency spectra will first be examined using basic signal analysis and the FFT.

A baseline time-frequency representation will then be produced using the STFT with overlapping Hann-windowed frames. Window length, hop size, and FFT size will be selected according to the required time-frequency resolution. Spectrograms will be used to examine how the spectral structure of the pouring sound changes as the container fills.

Initial features will include dominant frequency, spectral centroid, spectral spread, spectral roll-off, and RMS energy. These features were selected because they provide interpretable descriptions of spectral position, spectral distribution, and signal energy. The STFT-based system will act as a baseline rather than the final method. Suitable later ELEC5305 techniques, such as modulation analysis, auditory filter banks, adaptive time-frequency processing, and linear prediction, will be evaluated as possible extensions. Only techniques that are relevant to the pouring-sound problem will be included.

The estimation system will classify recordings or short time segments into several fill-level ranges, such as low, medium-low, medium-high, and high. A simple feature-based classifier, such as thresholding or k-nearest neighbours, may be used. Different signal representations and feature sets will then be compared using the same evaluation procedure.

## **6. Expected Outcomes**

The expected outcome is a working MATLAB prototype that accepts a pouring-audio recording and estimates the corresponding liquid fill-level category.

The project will produce spectrograms, extracted feature trajectories, and comparisons between different signal representations. Performance will be evaluated mainly using classification accuracy and a confusion matrix.

The main technical outcome will be a comparison showing whether later or more advanced signal representations provide measurable improvements over the initial STFT and spectral-feature baseline. MATLAB code, example results, documentation, and the final project material will be maintained in the GitHub repository.

## 7. Timeline

**Weeks 1–2:** Finalise the research question, project scope, and data source. Create the GitHub repository and review relevant literature.

**Weeks 3–5:** Download and inspect the selected dataset. Develop MATLAB preprocessing, FFT, STFT, spectrogram visualisation, and baseline feature extraction.

**Weeks 6–9:** Implement the initial fill-level estimator and evaluate the baseline system. Incorporate suitable new signal-processing techniques introduced during later ELEC5305 modules, such as modulation or auditory representations.

**Weeks 10–11:** Compare alternative representations and feature combinations, optimise parameters, calculate classification metrics, and analyse errors.

**Weeks 12–13:** Finalise experimental results, prepare the final report and figures, organise MATLAB code and documentation, and complete the GitHub project page.

## 8. References

- Bagad, P., Tapaswi, M., Snoek, C. G. M., & Zisserman, A. (2025). The sound of water: Inferring physical properties from pouring liquids. In *ICASSP 2025–2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 736–740). IEEE. <https://doi.org/10.1109/ICASSP49660.2025.10889950>
- Qian, X., Xu, J., Gao, Y., Li, M., Li, W., & Yin, X.-C. (2025). Understanding dynamic auditory and tactile perception for water filling level estimation. *International Journal of Social Robotics*, 17(10), 2097–2106. <https://doi.org/10.1007/s12369-024-01176-4>
- Schmelter, S., Fühner, C., & Röhrig, C. (2020). Container filling level estimation using vibration resonance behavior. In 2020 IEEE 5th International Symposium on Smart and Wireless Systems within the Conferences on Intelligent Data Acquisition and Advanced Computing Systems (IDAACS-SWS) (pp. 1–6). IEEE.

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